

Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MRBENCH | Context: Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)

Definition: Whether the benchmark’s output categories and scoring criteria reflect regionally valid decision boundaries.

Justification

MODERATE priority per elicitation. The eight dimensions and three-tier labeling [Q58, Q59] are accepted by the user as necessary and sufficient (elicitation Q3). However, two structural concerns affect deployment validity. First, DAMR weights all eight dimensions equally [Q64], whereas the user identified 'providing guidance' as the single most critical dimension; using DAMR directly as an acceptability proxy would dilute the signal Indian teachers care most about. Second, the 'providing guidance' rubric operationalizes guidance via Socratic scaffolding (hints, supporting questions, not revealing answers) [Q23, Q24], which the user (elicitation Q4) and field evidence [WEB-15, WEB-16] indicate may not match Indian direct-correction expectations — a structural validity concern about how the construct itself is operationalized. The decision rule for the most-critical dimension may encode a non-regional pedagogical value. Treating dimensions as orthogonal [Q30] is acknowledged as a simplification [Q116, Q120, Q121]. Tone label collapse to 'Non-offensive' [Q90] reaches 100% compliance, providing little discriminative signal.

ID	Evidence
WEB-15	Indian teachers face structural misalignment with Western-default AI outputs; value curriculum-aligned, direct guidance
WEB-16	MathTutorBench (related Western-default benchmark) shares same annotator-cultural-provenance pattern

Strengths

- Eight dimensions accepted as necessary and sufficient by the user (elicitation Q3) and pilot validation [Q132]
- Per-dimension DAMR scores reported separately [Q67], enabling manual re-weighting toward 'providing guidance' [WEB-19 confirms re-weighting feasible]
- Three-tier labeling [Q58, Q59] supports nuanced acceptability judgments rather than binary
- Desired labels are explicitly documented [Q65, Q128], making the decision rule auditable

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories are dimensionally relevant to the deployment — all eight are accepted by the user (elicitation Q3). However, the decision rule for 'providing guidance' favors Socratic scaffolding [Q23, Q24], which may not align with Indian direct-correction norms (elicitation Q4, [WEB-15]).

ID	Evidence
WEB-15	Indian teachers face structural misalignment with Western-default AI outputs; value curriculum-aligned, direct guidance

OO-2: No dimensions are flagged as missing per user elicitation (Q3). However, exam-readiness framing — culturally salient for Indian teachers per cultural notes — is not a separate dimension; it is implicitly subsumed under guidance/actionability without dedicated operationalization.

OO-3: The 'providing guidance' dimension's desired label encodes a Socratic/answer-withholding value [Q23 'a good tutor strategy is not to reveal the answer to the student immediately but rather provide helpful guidance']. This is a non-regional pedagogical assumption per elicitation Q4 and field evidence [WEB-15, WEB-16].

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OO-4: Stakeholder-driven taxonomy redesign is not required (user accepts dimensions) but rubric recalibration of the 'providing guidance' dimension and DAMR re-weighting are recommended.

OO-5: Structural validity concern: equal-weight DAMR [Q64] does not match the user's prioritization of 'providing guidance' as dominant. Content validity concern: the guidance dimension's decision rule may encode a Western pedagogical default [Q23, Q24] that diverges from Indian acceptability criteria. Both are documented limitations enabling targeted remediation.

Information Gaps

- Whether the 'providing guidance' rubric, when applied with Indian-specific definitions of acceptable guidance, would yield substantially different desired labels — requires Indian teacher rubric review
- No published weighted DAMR methodology calibrated to specific evaluator populations [WEB-19]

Requires Expert Verification

- Indian teacher review of the 'providing guidance' rubric definition [Q24] to determine whether direct-correction responses would be appropriately rewarded under current desired labels

Remediation

Gap 1: DAMR weights all eight dimensions equally [Q64], but 'providing guidance' is the user-identified dominant acceptability criterion, and the guidance rubric encodes Socratic-scaffolding values [Q23, Q24] that may not match Indian direct-correction norms.

Recommendation: Implement deployment-specific weighted DAMR with 'providing guidance' as dominant weight. Additionally, recalibrate the 'providing guidance' desired labels through expert review: define what constitutes acceptable direct-correction guidance for Indian teachers and add this as an alternative acceptable response pattern, rather than treating answer revelation as monolithically negative.

Gap 2: The benchmark does not aggregate to a binary acceptability decision and does not consider cumulative learning impact [Q120, Q121].

Recommendation: Define a deployment-specific decision rule that maps weighted DAMR (or per-dimension thresholds) to a binary 'acceptable / not acceptable' label aligned with the deployment's evaluator-role requirement. Document the threshold transparently and audit against Indian teacher binary judgments on a held-out set.